

Sixteenth Section. Numbers and functionals of an algebraic field.

§ 132. Definition of algebraic numbers.

An algebraic equation

$$F(x) = x^n + a_1x^{n-1} + \cdots + a_{n-1}x + a_n = 0 \quad (1)$$

whose coefficients are rational numbers will, for brevity, be called a rational equation. It has, as was shown earlier, at least one root and at most n distinct roots. The numerical approximation of such roots by rational numbers, decimals, continued fractions, and, where necessary, by adjoining the imaginary unit $i = \sqrt{-1}$, belongs to the analytic side already treated. Here the concern is not their numerical values but the arithmetical laws which follow from the defining equations.

A number θ satisfying a rational equation

$$F(\theta) = 0$$

is called an algebraic number. Rational equations thus provide algebraic numbers in arbitrary abundance. The question whether there are also non-algebraic numbers is postponed.

§ 133. Integral algebraic numbers.

An algebraic number θ is called an integral algebraic number if it satisfies an equation

$$\theta^m + A_1\theta^{m-1} + A_2\theta^{m-2} + \cdots + A_m = 0 \quad (1)$$

whose coefficients $A_1, \dots, A_m$ are rational integers. It is enough that among the rational equations satisfied by θ there is one of this kind.

The ordinary rational integers occur as special cases. Positive rational integers are also called natural numbers. The word “integer”, without qualification, will mean an integral algebraic number, rational or irrational.

1. An integral algebraic number which is rational is necessarily a rational integer. If $\theta = P/Q$ with P, Q coprime rational integers, then equation (1) forces every prime divisor of Q to divide P . Hence $Q = 1$.
2. The sum, difference, and product of two integers are again integers. If α, β satisfy

$$\alpha^\mu + a_1\alpha^{\mu-1} + \cdots + a_\mu = 0, \quad \beta^\nu + b_1\beta^{\nu-1} + \cdots + b_\nu = 0,$$

and if

$$\omega = \alpha + \beta, \quad \omega = \alpha - \beta, \quad \omega = \alpha\beta,$$

form the $\mu\nu$ quantities

$$\alpha^r\beta^s, \quad r = 0, 1, \dots, \mu - 1, \quad s = 0, 1, \dots, \nu - 1,$$

and denote them by $\omega_1, \dots, \omega_m$. The products $\omega\omega_i$ can be expressed linearly:

$$\omega\omega_r = c_{r,1}\omega_1 + c_{r,2}\omega_2 + \cdots + c_{r,m}\omega_m, \quad r = 1, \dots, m, \quad (2)$$

with rational integral coefficients. Eliminating $\omega_1, \dots, \omega_m$ gives

$$\det(c_{r,s} - \omega\delta_{r,s}) = 0, \quad (3)$$

an equation for ω with rational integral coefficients.

3. If $f(x)$ is irreducible over the rational field and one root of $f(x) = 0$ is integral, then all roots of $f(x) = 0$ are integral. Consequently the equation of least degree

$$f(x) = x^n + a_1x^{n-1} + \cdots + a_n \quad (4)$$

for an integral algebraic number has rational integral coefficients.

§ 134. Algebraic fields.

Let $f(x)$ be the irreducible equation of least degree satisfied by θ , and let

$$\theta, \theta_1, \dots, \theta_{n-1}$$

be its roots. These numbers are conjugate. The totality of rational functions of θ ,

$$\omega = \varphi(\theta),$$

with rational coefficients is a field. If R denotes the field of rational numbers, this field $R(\theta)$ is called an algebraic number field of degree n .

Every number of such a field has, in each conjugate field, a corresponding conjugate number. Symmetric functions of the conjugates are rational. Among them the most important are the sum and the product: the first is the trace and the second the norm,

$$S(\omega), \quad N(\omega).$$

Equal conjugates are counted with their multiplicities.

Since the four arithmetical operations can be carried out in an algebraic number field as in the rational field, the question arises how far the laws of arithmetic in the rational integers continue to hold there. The first issue is the decomposition into prime factors. This decomposition usually fails if one uses only the numbers of the field. The material of calculation must therefore be enlarged. Kummer did this for cyclotomic numbers by ideal prime factors, Dedekind by ideals, and Kronecker by a different general method. The following paragraphs connect these theories and develop, from Kronecker's starting point, what is needed for the later algebraic applications.

§ 135. Integral functions in an algebraic field.

Let Ω be an algebraic number field. A whole expression

$$\varphi(x, y, z, \dots) = \sum \alpha x^r y^s z^t \dots \quad (1)$$

with non-negative integral exponents and coefficients α in Ω is called an integral function in Ω . It is understood that equal monomials have been collected. Two such functions are equal only when corresponding coefficients are equal; one of them is zero only when all its coefficients are zero.

The degree of a term $x^r y^s z^t \dots$ is $r + s + t + \dots$, and the degree of a function is the highest degree of a term with non-zero coefficient. The degree of a product is the sum of the degrees of the factors.

The following facts from the first volume are used.

1. Integral functions in Ω are reducible or irreducible. Every reducible integral function is a finite product of irreducible integral functions, and the irreducible factors are determined by the function up to constant factors from Ω .
2. The product of two primitive integral functions in R is primitive. If

$$\Phi(x, y, z, \dots) = \sum a x^r y^s z^t \dots \quad (2)$$

has rational integral coefficients with greatest common divisor m , write

$$a_i = m e_i, \quad E(x, y, z, \dots) = \sum e_i x^r y^s z^t \dots \quad (3)$$

Then

$$\Phi = m E \quad (4)$$

and E is primitive.

3. The divisor-content of a product of integral functions is the product of the divisor-contents of its factors.

Replacing θ successively by its conjugates gives the conjugate functions

$$\varphi(\theta, x, y, \dots), \quad \varphi(\theta_1, x, y, \dots), \quad \varphi(\theta_2, x, y, \dots), \dots \quad (5)$$

Their product

$$N(\varphi) = \varphi\varphi_1\varphi_2\cdots \quad (6)$$

is an integral function in R , and

$$N(\varphi\psi) = N(\varphi)N(\psi). \quad (7)$$

§ 136. Functionals of an algebraic field Ω .

The variables used here are symbols of calculation rather than independent number-variables. We introduce integral and fractional functions in any number of variables with coefficients in Ω . Every such function can be written as

$$\omega = \frac{\varphi}{\psi}, \quad \psi \neq 0,$$

where φ, ψ are integral functions. Equality of two such fractions means

$$\varphi\psi_1 = \varphi_1\psi.$$

A fraction is irreducible when numerator and denominator have no common divisor. Such a function ω is called a functional of the field Ω . The functionals form a field, the functional field of Ω .

A functional with rational coefficients is rational. Every rational functional can be written as

$$A = a \frac{E_1}{E_2},$$

where a is a positive rational number and E_1, E_2 are primitive integral functions in R . Thus:

1. Every rational functional can be made an integral function in R by multiplication by a primitive function; several rational functionals can be written with a common primitive denominator.
2. The positive rational number a is the absolute value of A . For ordinary numbers this is the usual absolute value; the factor E_1/E_2 generalizes the sign.
3. The absolute value of a product of rational functionals is the product of their absolute values.

For a functional ω in Ω , $N(\omega)$ is a rational functional. The absolute value of $N(\omega)$ is the absolute norm:

$$N(\omega) = N_a(\omega)E(\omega). \quad (3)$$

Hence

$$N_a(\alpha\beta) = N_a(\alpha)N_a(\beta). \quad (4)$$

Replacing all coefficients by the corresponding numbers in a conjugate field gives a conjugate functional, and every equation between functionals remains true after this simultaneous replacement. If t is a new variable, then

$$N(t - \omega) = t^n + A_1t^{n-1} + A_2t^{n-2} + \cdots + A_n \quad (5)$$

has rational functionals as coefficients and vanishes for $t = \omega$. Among all such functions vanishing at ω , with leading coefficient 1, there is a unique one $F(t)$ of least degree. It is irreducible over R , every other such function is divisible by it, and

$$N(t - \omega) = F(t)^e.$$

Thus every functional is the root of a unique irreducible equation over R .

§ 137. Integral functionals.

A rational functional is integral when its absolute value is an integer. In general, a functional ω is integral when it satisfies

$$\omega^m + A_1\omega^{m-1} + A_2\omega^{m-2} + \cdots + A_m = 0 \quad (1)$$

with integral rational functionals as coefficients. If such a functional is rational, this definition agrees with the preceding one.

Let

$$\Phi(t) = t^m + A_1t^{m-1} + \cdots + A_m$$

have rational functionals as coefficients. Multiplication by a common denominator produces a primitive integral function

$$aE(x, y, \dots)\Phi(t) = P(t, x, y, \dots). \quad (2)$$

The primitive-factor theorem then gives: if an integral functional ω is a root of Φ , every rational factor of Φ has integral coefficients. In particular the irreducible equation of ω has integral coefficients.

Every functional becomes integral after multiplication by a rational integer, and therefore every functional is a quotient of two integral functionals.

A useful criterion is this. If there are quantities $\omega_1, \dots, \omega_m$, not all zero, such that

$$\omega\omega_i = A_{1,i}\omega_1 + A_{2,i}\omega_2 + \cdots + A_{m,i}\omega_m, \quad i = 1, \dots, m, \quad (5)$$

with integral rational functionals $A_{r,i}$, then ω is integral. Indeed,

$$\det(A_{r,i} - \omega\delta_{r,i}) = 0 \quad (6)$$

gives

$$\omega^m + A_1\omega^{m-1} + \cdots + A_m = 0. \quad (7)$$

The last coefficient is, up to sign,

$$(-1)^m A_m = \sum \pm A_{1,i_1} A_{2,i_2} \cdots A_{m,i_m}. \quad (8)$$

It follows that an integral function of integral functionals is an integral functional; in particular sums, differences, and products of integral functionals are integral. Conjugates of an integral functional are integral, and the absolute norm of an integral functional is a natural integer.

§ 138. Divisibility. Associated functionals. Units.

An integral functional α is divisible by another non-zero integral functional β if $\alpha : \beta$ is integral. Thus $\alpha = \beta\gamma$. The number 0 is divisible by every integral functional. The usual elementary rules follow:

$$\alpha, \alpha_1 \equiv 0 \pmod{\beta} \Rightarrow \alpha \pm \alpha_1 \equiv 0 \pmod{\beta},$$

and transitivity of divisibility is immediate.

Two integral functionals divisible by one another are associated. A functional associated with 1 is a unit. Products and quotients of units are units. If α is divisible by β , then

$$N_a(\alpha) = N_a(\beta)N_a(\gamma). \quad (1)$$

The absolute norm of a unit is 1. Conversely an integral functional of absolute norm 1 is a unit, since

$$\frac{1}{\varepsilon} = \frac{\varepsilon'}{N(\varepsilon)}.$$

Moreover $N(\alpha) = \alpha\alpha'$; hence the natural integer $N_a(\alpha)$ is divisible by α . There are therefore infinitely many natural integers divisible by a given integral functional, and if a is the least positive rational integer divisible by α , every rational integer divisible by α is divisible by a .

§ 139. Greatest common divisor.

Let $\alpha, \beta, \dots$ be finitely many non-zero integral functionals and let $x, y, \dots$ be new variables. Then

$$\delta = \alpha x + \beta y + \dots \quad (1)$$

is integral. If D is the absolute norm of δ , then

$$N(\delta) = D E(x, y, \dots) \quad (2)$$

with E primitive.

Substituting rational integers $x_0, y_0, \dots$ gives

$$\delta_0 = \alpha x_0 + \beta y_0 + \dots$$

The norm $N(\delta t - \delta_0)$, divided by the norm of δ , gives an integral equation for

$$\eta = \frac{\delta_0}{\delta}.$$

Thus η is integral and δ_0 is divisible by δ . In particular $\alpha, \beta, \dots$ are divisible by δ ; conversely δ is divisible by every common divisor.

Every functional associated with

$$\delta = \alpha x + \beta y + \dots$$

is therefore called a greatest common divisor of $\alpha, \beta, \dots$. If the greatest common divisor is a unit, the functionals are relatively prime.

Consequences:

1. If integral functionals $\xi, \eta, \dots$ exist such that

$$\alpha \xi + \beta \eta + \dots = \varepsilon$$

is a unit, then $\alpha, \beta, \dots$ have no common divisor.

2. If α is relatively prime to β and to γ , then it is relatively prime to $\beta\gamma$.
3. If α is relatively prime to β and $\alpha\mu$ is divisible by β , then μ is divisible by β .

§ 140. Prime functionals.

An integral functional π of Ω , not a unit, is called a prime functional if its only divisors are units and functionals associated with π . Otherwise it is composite.

For an integral ω and a prime π , either ω is relatively prime to π , or ω is divisible by π . Therefore:

1. If the product $\alpha\beta$ of two integral functionals is divisible by a prime functional π , then one of the factors is divisible by π .

By repeated application, if a product of several factors is divisible by π , at least one factor is divisible by π .

Let p be the least natural integer divisible by π . Then p is a rational prime, and the absolute norm of π is a power of p . Writing

$$p = \pi \omega \quad (1)$$

and taking norms gives

$$p^n = N_a(\pi) N_a(\omega). \quad (2)$$

Thus

$$N_a(\pi) = p^f, \quad (3)$$

where $1 \leq f \leq n$. The integer f is the degree of the prime functional π .

§ 141. Decomposition of integral and fractional functionals into prime factors.

Every non-zero integral functional which is not a unit is divisible by a prime functional. If it is not prime, write

$$\omega = \omega_1 \alpha, \quad (1)$$

where neither factor is a unit. Then

$$N_a(\omega_1) < N_a(\omega). \quad (2)$$

Repeating the process, the absolute norms strictly decrease, and the process must stop. Hence every such functional has a finite prime factorization:

$$\omega = \pi_1 \pi_2 \cdots \pi_\nu. \quad (5)$$

The factorization is unique up to associated prime functionals. If

$$\pi_1 \pi_2 \cdots \pi_\nu, \quad \chi_1 \chi_2 \cdots \chi_\mu \quad (6)$$

are associated products of primes, then χ_1 divides the first product, hence is associated with one of the π_i ; after cancelling, the argument repeats.

If a prime π occurs several times, the associated factors are collected into powers. We say that π occurs h times in ω if ω is divisible by π^h but not by π^{h+1} . Thus α is divisible by β precisely when every prime factor of β occurs in α at least as often.

For several functionals one may write

$$\varepsilon_1 \alpha = \pi_1^{a_1} \pi_2^{a_2} \cdots, \quad \varepsilon_2 \beta = \pi_1^{b_1} \pi_2^{b_2} \cdots. \quad (7)$$

The greatest common divisor is obtained by taking the smaller exponents; the least common multiple by taking the larger exponents. Fractional functionals are reduced by factorizing numerator and denominator and cancelling the greatest common divisor.

§ 142. Integral functions in the field Ω .

The preceding laws allow the proof of Gauss's lemma to be repeated for functionals. Let

$$\varphi = a_0 t^h + a_1 t^{h-1} + \cdots + a_h, \quad \psi = b_0 t^k + b_1 t^{k-1} + \cdots + b_k$$

be integral functions of one variable with integral functional coefficients. If their product

$$\chi = \gamma_0 t^{h+k} + \gamma_1 t^{h+k-1} + \cdots + \gamma_{h+k}$$

has all coefficients divisible by a prime functional π , then π divides all coefficients of φ or all coefficients of ψ .

Consequently, for

$$\varphi = \varphi_0 t^h + \varphi_1 t^{h-1} + \cdots + \varphi_h \quad (1)$$

the functional φ is integral only if all coefficients $\varphi_0, \dots, \varphi_h$ are integral. If not, introduce a principal denominator μ :

$$\chi = \mu \varphi = \alpha_0 t^h + \alpha_1 t^{h-1} + \cdots + \alpha_h. \quad (2)$$

A prime factor of μ which does not divide all α_i would contradict the preceding lemma when one uses an integrality equation

$$\varphi^m = c_1 \varphi^{m-1} + c_2 \varphi^{m-2} + \cdots + c_m \quad (3)$$

and multiplies through by powers of μ :

$$E\chi^m = \mu(a_1 E_1 \chi^{m-1} + a_2 E_2 \mu \chi^{m-2} + \cdots + a_m E_m \mu^{m-1}). \quad (4)$$

Repeated application gives: an integral rational function of several variables, whose coefficients are numbers or functionals involving other variables, is integral only when all its coefficients are integral.

Every integral functional is associated with an integral function whose coefficients are integers of Ω ; multiplication by a rational functional which is a unit transforms it into such a function.

§ 143. Prime factors of the numbers of Ω .

The numbers of Ω are among the functionals, so their factorization into primes exists, but the prime factors need not be numbers; in general they are functionals.

Equations between functionals are ultimately identities between integral rational functions. They remain true when variables are renamed. Therefore:

1. The prime factors of an integral functional ω may be represented so that they do not contain the variables on which ω depends.

For ω divides a number, for instance its absolute norm, and its prime factors are among the prime factors of that number.

If

$$E\omega = \varphi(x, y, \dots) \quad (1)$$

with E a unit and φ an integral function in Ω , then by replacing prime factors with variable-free representatives one gets a number associated with ω . Hence:

- 2 An integral function φ with integral number coefficients is the greatest common divisor of all its coefficients.

Now:

- 3 Given any integral functional ω , one may choose a number α divisible by ω such that $\alpha : \omega$ is relatively prime to any prescribed functional ρ .

First, for each prime functional π there is a number divisible by ω but not by $\omega\pi$. For finitely many distinct prime functionals $\pi_1, \pi_2, \dots$ one chooses such numbers and forms a sum divisible by ω but by none of the $\omega\pi_i$. Taking the prime functionals occurring in ρ gives the assertion.

It follows that:

- 4 Every integral functional ω of Ω is the greatest common divisor of two integral numbers and is associated with a binary linear form

$$\alpha x + \beta y$$

with integral numbers α, β .

Thus the main result of the section is that the integral algebraic numbers of a fixed field factor uniquely when the field is enlarged to its functional field by adjoining variables. The construction uses only the presence of integers in the ground field and their unique prime factorization. Hence the same reasoning applies to any ground field with these two properties, for example to the field of rational functions of one variable. This is the starting point for the theory of algebraic functions of one variable.